

Day 5 – AMI Creation & Launching New EC2 from Existing Server

Today's Topics

- What is AMI
 - Why AMI is required
 - Creating custom AMI
 - Launching EC2 from AMI
 - Understanding AWS Image Service
 - Server cloning concept in AWS
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Introduction to AMI

AMI stands for:

Amazon Machine Image

AMI is a template of an EC2 server.

It contains:

- Operating System
- Installed software
- Server configurations
- Applications
- Files and webpages

Using AMI, AWS can create multiple identical servers.

Real-Time Scenario

We already created:

- Ubuntu EC2 Server
- Installed NGINX
- Created custom webpage using index.html

Now we want:

- Same server again
- Same webpage again
- Same configurations again

Instead of repeating installation manually every time, AWS provides AMI service.

Existing Server Flow

Browser

↓

Public IP

↓

Security Group

↓

Ubuntu EC2

↓

NGINX

↓

index.html

Website Output:

Welcome Altham College Students

NGINX Server Running Successfully on Ubuntu EC2

What Happens During AMI Creation

AWS creates a template of the existing server.

AMI stores:

- Operating System
- Installed packages
- NGINX configuration
- HTML files

- Server settings
- Attached EBS snapshot

Flow:

Running EC2

↓

Create AMI

↓

AWS stores server image

↓

Launch identical EC2 anytime

Step 1 – Open EC2 Dashboard

Login to AWS Console.

Go to:

EC2 Dashboard

Select existing EC2 server:

ubuntu-nginx-server

Step 2 – Create AMI

Inside EC2:

Actions

→ Image and templates

→ Create Image

Step 3 – Configure Image Details

Example:

Setting	Value
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Image Name	nginx-custom-image
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Setting	Value
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Description	Ubuntu server with NGINX
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Click:

Create Image

AMI Creation Architecture

Ubuntu EC2



NGINX Installed



Custom Webpage



Create AMI



AMI Stored in AWS

Important Understanding

During AMI creation AWS:

- Takes EBS snapshot
- Copies server data
- Creates reusable server template

AMI includes:

- OS
 - Applications
 - Installed software
 - Configurations
 - Files
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Step 4 – Verify AMI

Go to:

EC2 → AMIs

Check image status:

Available

Step 5 – Launch New EC2 from AMI

Select:

nginx-custom-image

Click:

Launch Instance from AMI

Now AWS automatically creates:

- Same Ubuntu server
- Same NGINX setup
- Same webpage
- Same configurations

No need to install NGINX again.

EC2 Configuration

Setting	Value
Name	nginx-server-2
AMI	nginx-custom-image
Instance Type	t2.micro
Security Group	Allow SSH & HTTP
Storage	8 GB

Launch instance.

New Server Flow

Custom AMI



Launch EC2



Second Ubuntu Server



NGINX Already Installed



Website Already Running

Step 6 – Test Website

Copy second EC2 Public IP.

Open browser:

http://PUBLIC-IP

Expected Output:

Welcome Altham College Students

NGINX Server Running Successfully on Ubuntu EC2

Complete Architecture

First EC2 Server



Install NGINX



Create index.html



Create AMI



AWS Image Service



Launch New EC2



Second Server Ready

Why Companies Use AMI

Companies use AMI for:

- Fast deployment
 - Server backup
 - Auto Scaling
 - Disaster Recovery
 - Server cloning
 - Standardized environments
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Manual Installation vs AMI

Manual Setup

Install OS again

Install NGINX manually

Configure webpage again

Time consuming

Repeated effort

AMI Setup

Already included

Already installed

Already copied

Fast deployment

One-click setup

Useful Commands

Check NGINX status:

```
sudo systemctl status nginx
```

Restart NGINX:

```
sudo systemctl restart nginx
```

Check website files:

```
cd /var/www/html  
ls
```

Interview Questions

What is AMI?

AMI is a template used to create EC2 instances with preconfigured operating system and applications.

Why AMI is important?

AMI helps in fast deployment, backup, server cloning, and auto scaling.

Does AMI include installed software?

Yes. AMI includes operating system, applications, packages, and configurations.

Final Flow Summary

Create EC2

↓

Install NGINX

↓

Create Webpage

↓

Create AMI

↓

Launch New EC2 from AMI

↓

Same Website Available Automatically